LandGuard

Explainable Land Transaction Integrity & Fraud Detection Engine

LandTech 2026 · Track: Digital Land Records, Verification & Digitization · **Problem statement chosen: P3 — Fraud & Double-Selling Detection**

1. The problem in the Bangladesh context

Land disputes are among the most common civil cases in Bangladesh. Many trace back to records that disagree: a deed registered at the sub-registry office does not match the mutation recorded at the AC Land office; the same interest appears to be sold to more than one buyer; a seller transfers more than the share recorded in their name; or a parcel changes hands again within days. Each record can look valid on its own — the inconsistency only appears when the records are read *together*, which is slow and manual for an investigating officer.

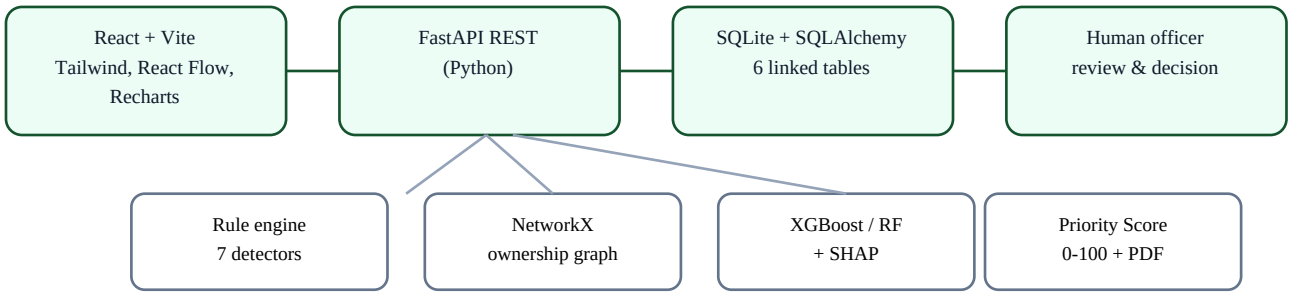
2. Proposed solution

Government systems maintain the records. **LandGuard adds an integrity-analysis layer** that checks whether linked records make sense together. It links parcel, owner, transaction, registration and mutation records into a time-aware ownership graph, runs seven deterministic checks, adds a machine-learning model to prioritise unusual cases, and shows plain-language evidence so an officer can decide where to look first.

Check	What it looks for
Double-sale	A seller transfers again after an earlier transaction already transferred their entire interest in the parcel (normal chains A→B→C are not flagged)
Over-transfer	Transferred share or area is larger than the seller's recorded share / controlled area
Rapid transfer	Parcel transferred again within 7 days (strong) or 8–30 days (moderate) — an anomaly signal, not proof
Ownership chain break	Recorded seller differs from the recorded previous owner
Registration / mutation mismatch	Missing record, or buyer, seller, area, type or date disagree between transaction, registration and mutation
Temporal inconsistency	Dates not in the order transaction ≤ registration ≤ mutation

Legal boundary. LandGuard is decision-support only. It never states that fraud is confirmed or who legally owns land. It outputs an **Investigation Priority Score (0–100)** — Low, Medium, High, Critical — not a fraud probability, and every alert ends in **Human Review Required**.

3. System architecture and technology stack



Layer	Technology (all open-source, runs offline on a laptop, no Docker)
Frontend	React, Vite, JavaScript, Tailwind CSS, React Router, Recharts, React Flow, Lucide icons
Backend	Python, FastAPI, Uvicorn, Pydantic, SQLAlchemy, Pandas, NumPy
Database	SQLite with primary keys, foreign keys and search indexes
Graph / ML / XAI	NetworkX; scikit-learn Random Forest and XGBoost; SHAP for explanations; joblib
Reports	ReportLab (PDF investigation report)

4. Key features

- **Parcel search** by district, upazila, mouza, khatian, dag or parcel ID; parcel profile with current owner and integrity status.
- **Ownership history** and **transaction timeline** rebuilt from the linked records.
- **Transaction graph** (React Flow): owner → transaction → owner, clickable, colour-coded for multiple transfers and conflicts.
- **Integrity alerts** with evidence tables, e.g. recorded share vs transferred amount and the potential excess.
- **Registration vs mutation comparison** side by side with mismatches highlighted.
- **AI analysis**: Investigation Priority Score with SHAP-based top contributing factors and a plain-language explanation.
- **Investigation report** (PDF) and a guided **judge demo** mode. Dashboard and analytics are computed live from the database.

5. Data, method and results so far

Data. Six synthetic datasets created by the team (no real personal data, no scraping): 15,000 parcels, 11,500 owners, 25,000 transactions, 23,370 registrations, 20,632 mutations and 25,000 label rows. A validation script checks IDs, foreign keys, dates, shares and areas before import.

Method. Rules give deterministic evidence (up to 70 points); an XGBoost model adds an anomaly score (up to 30 points). Labels are never used as model inputs, and stored ML scores are out-of-fold. Results on a stratified 80/20 test split (5,000 transactions):

Approach	Precision	Recall	F1
Rules only	95.3%	100.0%	97.6%
ML only (history/timing features)	94.2%	72.5%	81.9%
Hybrid (rules OR ML)	94.6%	100.0%	97.2%

Honest reading. The provided labels were generated from deterministic rules over the same records, so rules recover them almost perfectly and the hybrid does not beat rules-only here. The value of ML in this prototype is prioritisation and explanation; the ML-only row (no record-comparison features) is the fairer measure of what a model learns from patterns alone. Performance on real government data is untested and must be re-evaluated.

6. Plan for countrywide implementation

Phase	Scope
1. Pilot	One district: import registration (sub-registry) and mutation (AC Land) exports; officers review the prioritised list; collect feedback and false positives.
2. Integration	Read-only connection to e-Mutation / land-record systems; PostgreSQL for scale; login with roles and an audit log (JWT).
3. Scale	Division and national rollout, a graph database for cross-district chains, GIS / cadastral maps, Bangla interface.
4. Citizens	A 'check before you buy' view (P5) built on the same integrity checks.

Privacy and safety: access is for authorised officers; the system flags records for review and never changes or approves a registration or mutation.

7. Team and mentors

Role	Name	Affiliation / ID
Student 1 (Team Leader)		
Student 2		
Faculty mentor		
Industry mentor		

Fill in the names and DIU IDs before submission.

8. Limitations and AI tool disclosure

- Synthetic data and rule-derived labels (see section 5); no GIS; no login in this prototype.
- AI tools: Claude (Anthropic) was used as a coding assistant for code, tests and documentation, as disclosed in the repository README. Both student members must be able to explain the code and design.